

Table 1. Information-theoretic detectability bounds for fixed-key sampling methods.

Sampling family	TV bound	Inline meaning
Greedy	$\text{TV}(P^s, Q^{\text{greedy}}) = 1 - P^s(\mathbf{y}^*) = O(1)$	Q^{greedy} : point mass on $\mathbf{y}^*$
Biased (δ -tilt)	$\text{TV}(P^s, Q^{\text{bias}_\delta}) \leq \delta \sqrt{\frac{1}{4} \sum_{t=1}^T g_t (1 - g_t)} = O(\delta \sqrt{T})$	δ : bias strength; g_t : green-set mass
Bias-free (key E)	$\text{TV}(P^s, Q_E^{\text{bf}}) \leq \sqrt{\frac{1}{4} \sum_{t,v} \frac{\text{Var}_E[R_E(p_t)(v)]}{p_t(v)}} = O(\sqrt{T})$	E : key; R_E : reweighting operator
Semantic (key k)	$\text{TV}(P^s, Q_k^{\text{sem}}) \leq \sqrt{\frac{1}{2} \sum_{t=1}^{T_s} \mathbb{E} \left[\log \frac{1}{a_t} \right]} = O(\sqrt{T/\ell})$	a_t : acceptance mass; $T \approx \ell T_s$
Distribution-preserving	$\text{TV}(P^s, Q^{\text{prf}}) = 0$	$q_t \equiv p_t$

P^s denotes the baseline text distribution induced by standard stochastic sampling, and Q denotes the corresponding watermarked text distribution. T is the number of token-generation steps. In the semantic setting, T_s is the number of sentence-generation steps and ℓ is the average number of tokens per sentence, so typically $T \approx \ell T_s$.

In the greedy row, $\mathbf{y}^*$ is the deterministic greedy output sequence and Q^{greedy} is the point mass on that sequence.
In the biased row, δ is the logit-tilt strength, G_t is the keyed green set at step t , and $g_t := p_t(G_t)$ is the baseline probability mass assigned to that set.
In the bias-free row, E denotes the watermark key or seed, and R_E is the reweighting operator satisfying $\mathbb{E}_E[R_E(p_t)] = p_t$.
In the semantic row, a_t is the acceptance probability (acceptance mass) at sentence step t under the keyed semantic selection rule.
In the distribution-preserving row, $q_t \equiv p_t$ means that the conditional token distribution is preserved exactly at every generation step.
All bounds are stated for fixed-key constructions.